# 1. Monthly AQI figures that changed

These figures appear in the CAQM annual perspective tables titled “Month-wise Daily Avg. AQI for Delhi” (or “Month-wise Avg. AQI”). The later value is not automatically more correct; it is simply what the later publication prints.

| Same month and year | Earlier official release | Later official release | Difference | Source comparison |
|---|---|---|---|---|
| May 2021 | 146 (2022 release) | 144 (2023, 2024, 2025 releases) | 2 points | 2022 report \| 2023 report \| 2024 report \| 2025 report |
| September 2020 | 116 (2023, 2024 releases) | 118 (2025 release) | 2 points | 2023 report \| 2024 report \| 2025 report |
| October 2020 | 266 (2023, 2024 releases) | 265 (2025 release) | 1 point | 2023 report \| 2024 report \| 2025 report |
| November 2022 | 320 (2023, 2024 releases) | 321 (2025 release) | 1 point | 2023 report \| 2024 report \| 2025 report |
| April 2023 | 179 (2023, 2024 releases) | 180 (2025 release) | 1 point | 2023 report \| 2024 report \| 2025 report |

## What stands out

The 2021 May value is the clearest repeated mismatch: CAQM printed 146 in its 2022 perspective, then 144 in three later perspectives. Other small changes affect historical months in 2020, 2022, and 2023. The annual average AQI values shown in the 2023-2025 annual tables remain 2018: 225, 2019: 215, 2020: 185, 2021: 209, 2022: 209, 2023: 204, and 2024: 209. So the differences found here are in monthly figures; they do not change those rounded annual figures.

## A separate internal mismatch in CAQM's 2022 release

In the same CAQM 2022 perspective, the month-wise table lists December 2022 as 319. A later table in that release, “Comparative Daily Avg. Air Quality during Post-monsoon & Winter months,” lists December 2022 as 316. Both are presented as daily average AQI for Delhi. This is an internal table inconsistency in one official release. The source does not explain it. Open the 2022 release and check both tables.

## 2. Annual particulate-matter figures also shifted

CAQM perspectives and the CAQM Annual Report 2024-25 repeat annual average PM10 and PM2.5 values. The same historical year is not always printed with the same rounded concentration. Concentrations below are in micrograms per cubic metre (µg/m3); each pair is PM10 / PM2.5.

| Year | 2022 release | 2023 release | 2025 perspective | 2024-25 annual report |
|---|---|---|---|---|
| 2018 | 243 / 115 | 242 / 114 | 241 / 113 | 242 / 114 |
| 2019 | 218 / 108 | 217 / 108 | 217 / 107 | 218 / 109 |
| 2020 | 181 / 95 | 180 / 94 | 180 / 94 | 181 / 95 |
| 2021 | 213 / 105 | 210 / 104 | 211 / 105 | 211 / 105 |
| 2022 | 211 / 98 | 211 / 98 | 211 / 98 | 211 / 98 |
| 2023 | not listed | 205 / 100 | 205 / 100 | 206 / 100 |
| 2024 | not listed | not listed | 212 / 104 | 212 / 105 |

Values are PM10 / PM2.5 in µg/m3. The 2022, 2023, and 2025 columns come from their respective CAQM annual perspectives; the last column is the CAQM Annual Report 2024-25. Click to open: 2022, 2023, 2025, 2024-25 annual report.

The 2022 release provides the older series; later releases reprint earlier years and sometimes show different rounded annual concentrations. Some values go down and later go back up. That pattern is consistent with revisions or different aggregation, but the sources do not state the reason.

These changes are small but matter for reproducibility: a researcher downloading the official summary in a different year could transcribe a different historical value. They do not, by themselves, demonstrate bad sensor readings or a problem in your hourly data.

## 3. CPCB cross-check: Diwali 2021 matches CAQM

This is a useful control check where both agencies publish comparable values for the same event. CAQM's annual table gives Delhi AQI 382 on Diwali 2021 and 462 on the post-Diwali day. CPCB's daily bulletins show 382 on 4 November 2021 and 462 on 5 November 2021. The exact date/value pairs agree. CAQM 2023 perspective | CPCB bulletin, 4 Nov | CPCB bulletin, 5 Nov.

Therefore, this official-site search did not find evidence of a CAQM-vs-CPCB mismatch for this event. It did find the CAQM publication mismatches above.

### One time-period caution

CAQM's 2025 perspective describes its annual PM comparison as values through 29 December; a later CAQM/PIB release describes the 2025 annual particulate comparison through 31 December and gives different 2025 averages. Those periods are not identical, so this report does not count that as a like-for-like conflict. The later source is the 2026 PIB/CAQM release. Always compare the dates covered before flagging a difference.

# 4. What can responsibly be reported

## Supported conclusion

“CAQM has published different rounded historical monthly AQI values across its annual perspectives: for example, May 2021 is 146 in the 2022 release and 144 in the 2023, 2024, and 2025 releases. Small historical changes are also visible for September/October 2020, November 2022, and April 2023. CAQM's 2022 report also gives two December 2022 values (319 and 316) in separate tables. I have not established why these values differ, or which should be considered final.”

## Avoid saying

Do not say “CPCB/CAQM hourly data is wrong” based on this report. The evidence here is from published summary figures, mostly monthly or annual, and does not compare the agencies' full hourly files. The report also does not use project-local values.

## Questions to send to CAQM

1. For May 2021, should researchers use 146 or 144 as the official monthly average AQI? Was one figure revised, and where is the correction documented?
2. Were the historical monthly AQI values for September/October 2020, November 2022, and April 2023 recalculated after the earlier annual perspectives?
3. In the 2022 perspective, why is December 2022 319 in the month-wise table but 316 in the post-monsoon/winter table?
4. Which source dataset, averaging procedure, and rounding rule should researchers cite for the final historical summary?

## Interpretation

Because the AQI differences are only 1-2 points and the annual AQI is unchanged, the most cautious explanation is that the historical tables may have been updated or rounded differently. That is an inference, not an explanation stated by CAQM. The annual PM differences are also small and could reflect revisions or changed aggregation. The public documents do not settle this.

# 5. Official source register

Every link below is an official Government of India/CAQM/CPCB publication used in this comparison. Click a source title to open it.

- CAQM 2022 - Air Quality Perspective in Delhi; PIB release dated 2 Jan 2023.
- CAQM 2023 - Air Quality Perspective in Delhi; PIB release dated 31 Dec 2023.
- CAQM 2024 - Air Quality Perspective in Delhi; PIB release dated 31 Dec 2024.
- CAQM 2025 - Air Quality Perspective in Delhi; PIB release dated 31 Dec 2025.
- CAQM Annual Report 2024-25; Delhi air-quality section.
- CAQM 2025 annual particulate summary; PIB release dated 2 Jan 2026.
- CPCB Air Quality Status for Delhi & NCR, dated 4 Nov 2021.
- CPCB Air Quality Status for Delhi & NCR, dated 5 Nov 2021.

## Scope and limitation

The comparison covers CAQM annual perspective tables for 2022-2025, the CAQM Annual Report 2024-25, and two CPCB event bulletins from November 2021. The historical data reviewed spans 2018-2025. This is a targeted audit of accessible official summaries, not a complete audit of every CPCB/CAQM bulletin, station file, or hourly observation. A confirmed publication mismatch should be sent to the agencies as a clarification request, not presented as proof of misconduct or incorrect raw readings.

Prepared for the Delhi AQI forecasting project. Values are quoted as published, without recomputing averages. Source pages were reviewed on 25 October 2026.